

AIR TRAFFIC CONTROL SPEECH SYNTHESIS WITH MODEL FINE-TUNING AND ITS SECURITY IMPLICATIONS

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ABSTRACT

Analog very high frequency (VHF) voice links in air traffic control (ATC) lack cryptographic source authentication. Public text-to-speech models can generate controller-conditioned speech for potential unauthorized instructions. We compare pretrained Chatterbox with versions fine-tuned on recordings from other controllers. Across 270 messages from 9 controllers at the 6-s reference limit, adaptation changes log-duration error by -0.0017 and word error rate (WER) by $+0.684$ percentage points, without confirming improved duration matching within a 1-percentage-point WER tolerance. On 90 matched messages, base synthesis yields 14.28% WER with 2-s references. Fine-tuning increases speaker-embedding similarity at 2- and 4-s limits, without confirmed improvement at 6 s. Predicted naturalness declines, and pitch-slope error increases on its measured subset. Public pretrained models already provide controller-voice conditioning, lowering the technical barrier to hypothetical voice spoofing. The tested adaptation establishes no consistent advantage.

Index Terms— Air traffic control, aviation security, text-to-speech synthesis, speaker similarity, domain adaptation

1. INTRODUCTION

Air traffic control (ATC) depends on correctly understood instructions from authorized sources. The 1977 Tenerife collision killed 583 people after the KLM aircraft began takeoff without clearance. The investigation identified overlapping radio transmissions and inadequate language among the contributing factors [1].

In Roanoke in 1993, an individual impersonating an air traffic controller issued false flight instructions that one pilot began to follow. Legitimate controllers corrected the misinformation [2]. Legacy analog very high frequency (VHF) voice lacks cryptographic source authentication [3]. Procedural defenses include pilot acknowledgment, controller checking of readbacks, and prescribed responses to false communications [4].

Speaker-authorization research uses received-radio voices to identify unauthorized speakers [5], a task that voice-conditioned text-to-speech (TTS) could complicate. Chatterbox provides an open-source TTS package and pretrained weights for generating speech from text and reference audio in a single inference call [6]. Our earlier Integrated Communications, Navigation and Surveillance (ICNS) study used pretrained Chatterbox with public AT-COSIM recordings and reported strong word recovery and measurable speaker resemblance, alongside less consistent prosody [7]. The

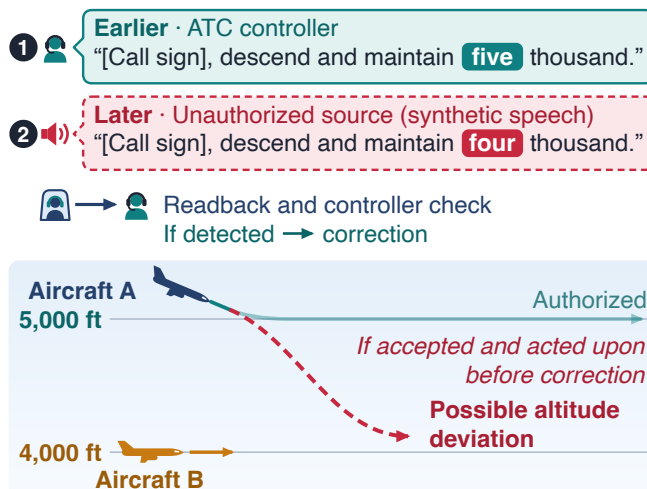


Fig. 1. Conceptual ATC threat scenario. A later unauthorized deepfake message changes the requested altitude from 5,000 to 4,000 feet. The dashed trajectory assumes acceptance and action before correction. Pilot readback to the legitimate controller provides an opportunity for detection. Aircraft positions are schematic, and a traffic conflict may occur depending on geometry and timing.

speech-generation capability for hypothetical controller deepfakes is therefore already available without custom model training.

Aviation TTS research also addresses controller training [8] and synthetic data generation for speech recognition [9]. Building on the ICNS findings, we test whether controller-excluded fine-tuning adds value beyond pretrained synthesis with brief references. Matched text, references, and generation seeds separate the adaptation effect from reference conditioning. A joint duration and word error rate (WER) criterion tests improved duration matching within a word-error tolerance.

2. THREAT MODEL

2.1. Voice spoofing scenario

An unauthorized VHF instruction may sound plausible enough for a pilot to act on it before its source is challenged. In Melbourne in 2016, a Boeing 737 crew received an unauthorized go-around instruction at about 200 feet above ground and initiated the maneuver, judging that questioning it could add confusion. The controller denied issuing the instruction as the maneuver began, and the aircraft

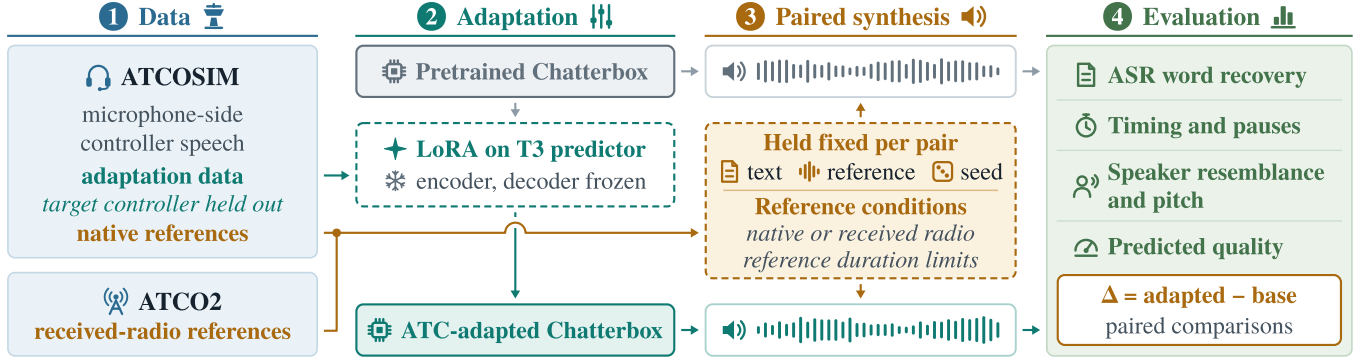


Fig. 2. Study design. ATCOSIM [10] supplies recordings for TTS fine-tuning, audio samples for voice conditioning, and held-out speech and transcripts for evaluation. A separate comparison uses received-radio recordings from ATCO2 [11] for voice conditioning while retaining ATCOSIM message text. Each native low-rank adaptation (LoRA) model is fine-tuned on controllers other than the evaluation speaker. Base and adapted models generate the same messages using identical conditioning audio and generation seeds. Automatic speech recognition (ASR) measures word recovery. Paired differences are $\Delta = \text{adapted} - \text{base}$.

later landed safely [12]. This incident does not establish the use of voice deepfakes.

A hypothetical adversary may attempt voice spoofing with speech synthesized to resemble a controller, termed a voice deepfake here. The scenario assumes a controller recording, a TTS system, and enough context to compose a plausible instruction, with unauthorized VHF transmission as the attack vector.

Figure 1 depicts an acknowledged authorized clearance followed by an unauthorized message mistaken for an altitude revision. The false instruction follows Federal Aviation Administration (FAA) altitude phraseology [4] and takes the form of an amended clearance, which is part of normal operations [13]. The scenario assumes reception of the unauthorized transmission and allows for controller intervention upon hearing either the message or the pilot’s readback.

2.2. Operational defenses and study scope

The FAA requires acknowledgment of instructions and checking of items read back. Its false-communication procedure calls for correction of information and alerts to affected aircraft [4]. FAA guidance also recommends altitude readback for mutual verification [13]. In Melbourne, the investigation found minimal overall operational impact from the unlawful broadcasts [12].

Aircraft identity, current clearance, traffic context, and controller response also influence a message’s credibility. We evaluate the speech-generation component using unchanged corpus transcripts. WER compares recognized outputs with the text supplied at TTS inference. Pilot acceptance, aircraft response, over-the-air transmission, and correction timing are outside this evaluation. The measured voice and delivery features characterize synthesis rather than successful spoofing.

3. PROBLEM FORMULATION

3.1. ATC language and controller delivery

A speech register is the language and delivery associated with an activity. ATC phraseology uses concise instruction patterns, aircraft identifiers, and numerical values [4]. A call sign identifies the aircraft addressed, and an altitude, heading, or frequency specifies an action or setting. Changing a value can change the required action even when the surrounding phraseology remains familiar.

Trippie and Baese-Berk found that American controller speech had a narrower pitch range, faster articulation, and different vowel and consonant timing than professional radio broadcasters’ speech [14]. ATCOSIM includes English speech from German and Swiss controllers whose first languages are German, Swiss German, and Swiss French [10]. These findings motivate our pitch and timing measures, but we assess delivery against ATCOSIM recordings to reflect the speaker population represented in this study.

3.2. Controller-conditioned speech synthesis

For transcript x_i , a separate reference recording r_c of controller c , and a fixed generation seed z_i , the base model produces $\hat{y}_{ci}^0 = G_{\theta_0}(x_i, r_c, z_i)$. The adapted model replaces θ_0 with θ_c , learned without recordings from controller c . We compare each output with the recorded utterance y_{ci} and its transcript to assess word recovery, temporal correspondence, voice resemblance, and acoustic changes.

Reference conditioning supplies target-voice information at inference. Fine-tuning uses other controllers’ speech to adapt the model to the ATC register. Fixed text and references isolate the adapter’s contribution to duration matching within a word-error bound. Controller exclusion applies to our fine-tuning data. Base-model pretraining exposure is incompletely known.

4. MODEL FINE-TUNING

4.1. ATC data and controller separation

ATCOSIM contains non-prompted speech recorded at controller headset microphones during real-time simulations [10], upstream of any VHF transmission. Of its 10 controllers, 9 contribute 90 development, 180 validation, and 270 evaluation utterances. An author checked every evaluation transcript against its full recording.

Screening yields 5,881 recordings totaling 6.39 hours. We withhold development, validation, reference, and evaluation recordings and equivalent transcripts from training, and each native fit excludes all recordings of the evaluation reference speaker. Reference and evaluation recordings come from different sessions for 8 controllers and from a shared session, at least 60 s apart, for the remaining controller. Training sessions may overlap other partitions.

4.2. Model adaptation and fine-tuning

Chatterbox predicts speech tokens from text and a voice reference, then decodes them into a waveform [6]. We apply low-rank adaptation (LoRA) [15] to 120 attention projections in its T3 speech-token predictor, leaving the pretrained weights, speaker encoder, and S3Gen waveform decoder fixed. Rank-16 adapters use scaling $\alpha = 32$ and dropout 0.05, giving 3,932,160 trainable parameters.

Training minimizes token-normalized cross-entropy over speech and end-of-sequence tokens, supervising both token prediction and utterance termination. WER, duration error, speaker cosine, and predicted quality remain evaluation measures and do not enter the loss directly. Each training reference is a separate recording of the training utterance’s speaker with a different transcript. Assigned reference durations span 1–6 s and remain fixed across training seeds.

Training uses AdamW with batches of 20 utterances. The learning rate rises linearly to 5×10^{-5} at update 33, then remains constant. Selection evaluates checkpoints at 16, 33, 66, 132, 264, and 528 updates on held-out development messages under the first training seed. Eligibility requires complete measurements, reduced mean duration error, and a WER upper bound below tolerance. The eligible checkpoint with lowest duration error uses 33 updates and 660 distinct training utterances per fit. Longer fits fail the WER condition. This duration is fixed before confirmation evaluation across 3 training seeds.

4.3. Inference controls and repeated generation

We hold Chatterbox’s public defaults [6] fixed across models and reference lengths to compare adaptation without separately optimizing generation settings. Temperature 0.8 rescales token logits, classifier-free guidance 0.5 weights the contrast between predictions with and without text conditioning, and exaggeration 0.5 sets emotion conditioning. A repetition penalty of 1.2 downweights previously sampled speech tokens. Min- $p = 0.05$ filters candidates relative to the highest token probability, and top- $p = 1.0$ disables additional nucleus filtering. Generation stops at the end-of-sequence token or the native 1,000-token cap, and S3Gen retains 10 flow-matching steps.

Native inference uses 2 references per controller, cropped at reviewed word boundaries within 2-, 4-, and 6-s limits. Nested crops preserve original samples but vary duration and phonetic content together. We cache reference conditioning for repeated generation and reset a message-specific seed before each output, sharing it across model and reference conditions. For each message and reference limit, the 2 references yield 2 base outputs and 6 adapted outputs across 3 training seeds. Outputs retain watermarking at 24 kHz.

5. EVALUATION EXPERIMENTS

5.1. Recordings and matched comparisons

The primary native panel uses 270 messages at the 6-s reference limit, with 20–50 messages per controller. A matched 90-message panel, 10 per controller, covers the 2-, 4-, and 6-s limits (Fig. 2). A separate 8-s comparison covers 50 messages from 5 controllers with 1 reference each. Its speech prompt remains at 6 s, but speaker and waveform conditioning receive the longer crop [6].

Received-radio conditioning uses 10 controller-labeled ATCO2 recordings from different session groups [11], which need not represent distinct speakers. Each recording conditions 5 ATCOSIM texts, giving 50 cases at each 2-, 4-, and 6-s limit, and the adapter uses all 9

eligible ATCOSIM controllers. Generated outputs receive no radio-channel processing, and matched microphone-side ATCO2 recordings are unavailable.

5.2. Evaluation metrics and statistical analysis

WER measures recovered text, including call signs and instruction values, using Whisper-ATC transcripts [16] with digitwise numeral expansion and “niner”/“nine” normalization. Errors receive equal weight, although Trippe distinguished phraseology-consistent omissions from incorrect numerical and nonnumerical repetitions [17]. Command understanding requires evaluation beyond WER [18].

Duration error $e_D = |\ln(\hat{D}/D)|$ uses synthesized ($\hat{D}$) and recorded (D) durations including pauses, with 0 for equal duration and equal penalties for doubling or halving it. Pause-fraction error compares internal-silence proportions, but neither measure locates words or pauses within instructions. Envelope modulation characterizes speech dynamics at syllabic timescales [19]. We define envelope distance as 1 minus cosine similarity between log-energy modulation power spectra averaged over matched counts of central segments, using a study-defined 2–12 Hz band. pYIN [20] supplies pitch measurements for messages with valid values in every compared condition. Pitch-slope error is the absolute difference between median pairwise semitone slopes after normalizing each waveform’s time to unit duration.

ECAPA-TDNN speaker-embedding cosine [21] compares native outputs with recorded targets and radio-conditioned outputs with a fixed longest approved crop. Genuine same-speaker comparisons help interpret speaker-embedding scores [22]. Predicted naturalness and quality come from UTMOSv2 [23] and SCOREQ [24], respectively.

Outputs are averaged within messages and then controllers, with equal controller weights. Conditional 95% intervals use 2,000 paired within-controller message bootstrap resamples. The primary criterion requires upper bounds $U_D < 0$ and $U_W < 0.01$ for adapted-minus-base duration-error and WER differences, with WER expressed as a fraction. The 1-percentage-point tolerance is study-defined, without operational safety validation. Intervals condition on tested controllers, references, trained models, and generation draws. Shared training data and sessions limit population generalization. Radio comparisons weight recording groups equally. Secondary intervals are unadjusted for multiple comparisons.

6. RESULTS AND SECURITY IMPLICATIONS

6.1. Off-the-shelf synthesis with brief references

The base model produces ATC speech from brief references, with higher mean speaker cosine as references lengthen.¹ On the matched 90-message panel, base-model WER is 14.28%, 14.54%, and 14.80% at 2-, 4-, and 6-s reference limits. Mean speaker cosine is 0.4993, 0.5560, and 0.5805, respectively, with actual reference lengths averaging 1.777, 3.793, and 5.776 s. These outputs use unchanged pretrained weights.

Adaptation increases mean speaker cosine by 0.0139 at 2 s and 0.0098 at 4 s, with conditional 95% intervals [0.0070, 0.0212] and [0.0025, 0.0174], respectively. Adapted scores nonetheless remain below base synthesis with 6-s references and, on a separate subset, 8-s references. At 2 s, WER increases by 1.488 percentage points and log-duration error by 0.022. Insufficient 2-s pitch-range support limits the delivery comparison.

¹Audio companion: <https://daonvu.github.io/icassp-2027-atc-tts/>.

Table 1. Native comparison at 6-s reference limits with equal controller weights. $n = 270$ messages with common measurement support, except pitch ($n = 162$). Arrows indicate preferred measurement directions. Bold values mark the better point estimate. Primary outcomes appear above the rule.

Measure	Base	Adapted
Whisper-ATC WER (%) ↓	14.491	15.175
Log-duration error ↓	0.2122	0.2105
Pause-fraction error ↓	0.0479	0.0424
Envelope distance ↓	0.2777	0.2762
Pitch-median error (semitones) ↓	1.647	1.636
Pitch-range error (semitones) ↓	2.786	2.957
Pitch-slope error (semitones/utterance) ↓	7.501	7.872
Speaker cosine ↑	0.5803	0.5796
UTMOSv2 ↑	2.749	2.688
SCOREQ ↑	3.530	3.510

6.2. Effects of ATC fine-tuning

At 6-s references (Table 1), log-duration error changes by -0.0017 , with conditional 95% interval $[-0.0149, 0.0123]$, and WER by 0.684 percentage points, with interval $[-0.001, 1.410]$. Both intervals include 0, and the WER upper bound exceeds the predefined 1-percentage-point tolerance, so the joint criterion is not met. These intervals also do not establish model equivalence.

Recorded speech has mean WER of 17.597% and UTMOSv2 of 2.264, compared with 14.491% and 2.749 for base synthesis. These are recognizer and prediction-model scores, not listener ratings.

Fine-tuning reduces mean output duration from 4.408 to 4.030 s. Mean excess over recorded duration falls from 0.537 to 0.328 s, and mean shortfall rises from 0.398 to 0.566 s. All waveforms contribute to these means, with 0 for absent excess or shortfall. Shortening can improve overly long outputs and worsen short ones.

Across all 270 targets, mean pause-fraction error changes by $\Delta = -0.0055$. A descriptive split shows reduced error for the 174 targets without detected pauses and increased error for the other 96.

UTMOSv2 changes by -0.062 , indicating lower predicted naturalness, and pitch-slope error increases by 0.371 semitones/utterance on the 162-message subset. Pitch support is uneven, including only 2 of 20 and 7 of 50 messages for 2 equally weighted controllers. Changes in envelope distance, pitch-median and pitch-range errors, 6-s speaker cosine, and SCOREQ remain uncertain because all corresponding secondary intervals include 0.

A second ATC recognizer [25] also leaves the WER tolerance unconfirmed, yielding 25.726% base and 26.397% adapted WER with a change interval that includes 0 and extends above tolerance. A word-count-weighted sensitivity analysis shows higher WER after adaptation, without replacing the primary result.

6.3. Received-radio reference conditioning

Only the 4-s radio-reference cosine increase, 0.021, has an interval excluding 0. Cosine changes at 2 and 6 s and WER changes at all 3 limits remain uncertain. Because the comparison changes corpus, adaptation population, and similarity reference, its scores cannot rank native and radio-conditioned voices directly.

Table 2. Exploratory mean speaker cosine against 90 separate development recordings previously used in checkpoint selection. Reference comparisons differ from Table 1.

Query speech	Same controller	Other controllers
Genuine	0.692	0.097
Base TTS	0.559	0.095
Adapted TTS	0.558	0.096

6.4. Security interpretation and limitations

For both models, mean cosine is higher for the intended controller than for other controllers, but lower than for genuine same-controller speech (Table 2). Against this separate recording bank, the 6-s adaptation change is uncertain at -0.0013 .

Our ICNS study reported 3.9% mean WER and 0.726 mean speaker cosine across 1,000 ATCOSIM utterances [7]. Its Whisper large-v2 recognizer was fine-tuned on ATCO2 and ATCOSIM, whereas the large-v3 variant used here was fine-tuned on ATCO2 [16]. The WER values are therefore not directly comparable. It also found weak pitch-contour matching and measurable cadence deviations. Together with the present base-model results, this supports an accessible controller-deepfake capability through pretrained synthesis and public reference speech. The tested adaptation offers no confirmed overall advantage, even though some short-reference cosine scores improve.

Familiar phraseology and voice resemblance could lend credibility to a spoofed instruction, but the operational outcome depends on the context and defenses described in Section 2. Reference limits characterize the tested samples, not a minimum acquisition time. Model and recognizer pretraining exposure is uncertain, and isolating register effects requires matched non-ATC training controls.

7. CONCLUSION

Off-the-shelf Chatterbox already supports hypothetical controller deepfakes using pretrained weights, message text, and a brief controller recording. This extends our ICNS findings on word recovery and speaker resemblance [7] by showing that the tested ATC fine-tuning adds no confirmed overall advantage under matched conditions. At 6 s, adaptation did not meet the joint duration and word-error criterion. Speaker cosine increased at 2- and 4-s limits, but remained below base synthesis with 6-s references.

Adaptation changed delivery unevenly. Shorter outputs reduced excess duration but increased shortfall. Pause-fraction error fell mainly for targets without detected pauses. Predicted naturalness declined, and pitch-slope error increased on the measured subset. At 2 s, higher cosine accompanied higher word and duration errors. Greater embedding similarity can therefore coexist with poorer word recovery and delivery.

Future work should combine speech analysis and instruction extraction with trusted clearance records, surveillance data, and geographic constraints. Listening and simulator studies could compare genuine, base, and adapted speech to assess unauthorized-instruction detection, false alarms, comprehension, and correction timing. These evaluations should include unmodified public models, which already provide controller-voice conditioning.

8. COMPLIANCE WITH ETHICAL STANDARDS

This study used existing ATCOSIM and ATCO2 recordings without recruiting new participants. Synthesis used corpus transcripts and retained the model's watermarking. Evaluation used stored recordings without operational transmissions. The authors declare no conflicts of interest.

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